Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

SEC: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **JEE-MAIN** **Date: 07-01-2026**
Time: 09.00Am to 12.00Pm **GTM-10** **Max. Marks: 300**

KEY SHEET

MATHEMATICS

1	**3**	2	**2**	3	**4**	4	**1**	5	**3**
6	**3**	7	**4**	8	**4**	9	**2**	10	**3**
11	**4**	12	**2**	13	**3**	14	**3**	15	**2**
16	**3**	17	**2**	18	**1**	19	**3**	20	**3**
21	**0**	22	**2**	23	**6**	24	**3**	25	**0**

PHYSICS

26	**2**	27	**4**	28	**1**	29	**2**	30	**3**
31	**2**	32	**4**	33	**2**	34	**1**	35	**1**
36	**3**	37	**3**	38	**3**	39	**3**	40	**2**
41	**3**	42	**3**	43	**1**	44	**3**	45	**3**
46	**250**	47	**0**	48	**135**	49	**0**	50	**5**

CHEMISTRY

51	**1**	52	**3**	53	**1**	54	**2**	55	**1**
56	**4**	57	**4**	58	**4**	59	**1**	60	**1**
61	**2**	62	**4**	63	**1**	64	**1**	65	**1**
66	**1**	67	**1**	68	**3**	69	**3**	70	**3**
71	**1**	72	**3**	73	**33**	74	**5**	75	**45**

For More Join: @IITJEE_Advanced

SOLUTION
MATHEMATICS

1. If A is $n\times n$ matrix then $|adj\ A|=|A|^{(n-1)}$

2. Passing points of lines L_1 & L_2 are $(\lambda,2,1)$ &$\left(\sqrt{3},1,2\right)$

$$S.D=\frac{\begin{vmatrix}\sqrt{3}-\lambda & -1 & 1\\ -2 & 1 & 1\\ 1 & -2 & 1\end{vmatrix}}{\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ -2 & 1 & 1\\ 1 & -2 & 1\end{vmatrix}}$$

$$1=\left|\frac{\sqrt{3}-\lambda}{\sqrt{3}}\right|$$

$$\lambda=0,\lambda=2\sqrt{3}$$

3. Put $xe^x=t$

4. required area $2\int_0^1\left(\frac{y^2+3}{4}-\frac{y^2+1}{2}\right)dy=\frac{1}{3}$

05. $\lim\left(\cos x+a^3\sin\left(b^6x\right)\right)^{1/x}$

$$=e^{\lim\limits_{x\to0}\frac{\cos x+a^3\sin b^6x-1}{x}}$$

$$e^{\lim\limits_{x\to0}\left(\frac{\cos x}{x}+\frac{a^3\sin b^6x-1}{x}\right)}$$

$$e^{0+a^3b^6}=e^{a^3b^6}$$

$$a^3b^6=512$$

$$ab^2=8$$

06. $\overline{\beta}=\overline{\beta}_1-\overline{\beta}_2$

$$\overline{\beta}=\lambda\overline{\alpha}-\overline{\beta}_2$$

By using $\overline{\alpha}.\overline{\beta}=0\Rightarrow\lambda=\frac{1}{2}$

Cross product with $\overline{\beta}_1$ for 1

$$\overline{\beta}_1\times\overline{\beta}_2=\frac{1}{2}(-3i+9j+5k)$$

7. Given $\frac{x^2}{a^2}-\frac{y^2}{1}=1\ and\ \frac{x^2}{4}+\frac{y^2}{3}=1$

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Form the given canditial $\frac{2}{a}=\frac{2.3}{2}\Rightarrow a=2/3$

$1=\frac{4}{9}\left(e_n^2-1\right)\Rightarrow e_n^2=\frac{13}{4}\,12\left(\frac{13}{4}+\frac{1}{4}\right)=42$

$\frac{3}{4}=1-e_E^2\Rightarrow e_E^2=\frac{1}{4}$

8. $\ln|x|=-at+c$

$\ln|y|=-bt+\lambda$

$3y(1)=2x(1)\Rightarrow 3e^b=2\left(2e^{-a}\right)\Rightarrow e^{a-b}=\frac{4}{3}$

Form x(t) =y(t)

$\Rightarrow 2e^{-at}=e^{-bt}\Rightarrow 2-e^{ta-ba}\Rightarrow 2=\left(\frac{4}{3}\right)^t\Rightarrow \log_{\frac{4}{3}}2=t$

9. Given function is

$$f(x)=\begin{cases}0 & ;x<0\\ ae^x-1 & ;0\le x<1\\ b-1 & ;1<x<2\\ -c & ;\ x\ge 2\end{cases}$$

F(x) is discontinuous at x=1 because $ae-1\ b\ \&\ (b-1)$ cannot be equal

For continuous at x = 0 then $a-1=0\Rightarrow a=1$

For continuous at x = 2 then, $b-1=-c\Rightarrow b+c$

Here f is discontinuous only at point then a + b+ c = 2

10. Putting $x=i$, we get $\left(1+i+i^2\right)^{48}=a_0+a_1i+a_2i^2+.....+a^{98}i^{96}$

$\Rightarrow i^{48}=\left(a_0-a_2+a_4-.....\right)+i\left(a_1-a_3+.....\right)$

Equating the real parts we get

$a_0-a_2+a_4-a_6+.....+a_{96}=1$

11. **conceptual**

12. Given,

7 boys and 5 Girls are to be seated around a circular such that no two girls to be seated together, now we know that n objects can be arranged in a circle in $(n-1)!$ ways.

Let us first arrange 7 boys in circular arrangement in $(7-1)!$ ways.

Now there will be 7 gaps.

So let us select any 5 gaps out of 7 gaps and arrange 5 girls in the chosen gaps. This can be done in ${}^7C_5\times 5!$ ways.

Hence, required arrangements are $6!\times {}^7C_5\times 5!$

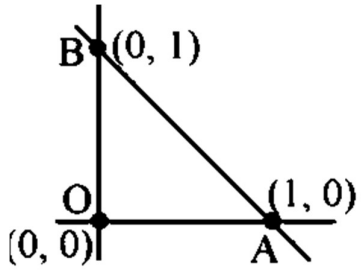
$=6\times 5!\times\frac{7\times 6}{2}\times 5!$

$=126(5!)^2$

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Therefore, required arrangements are $=126(5!)^2$

13. $(2k,3k)$ Will lie on circle whose diameter is AB



$(x-1)(x)+(y-1)(y)=0$

$x^2+y^2-x-y=0.....(i)$

Satisfying $(2k,3k)$ in (i)

$(2k)^2+(3k)^2-2k-3k=0$

$13k^2-5k=0$

$k=0,k=\frac{5}{13}$

Hence $k=\frac{5}{13}$

14. Hint: Here use concept $f(x)+f(1-x)=1$

Sol: $f(n)+f(1-n)=\frac{3^n}{3^n+\sqrt{3}}+\frac{3^{1-n}}{3^{1-n}+\sqrt{3}}$

$=\frac{3^n}{3^n+\sqrt{3}}+\frac{3}{3+\sqrt{3}.3^n}$

$=\frac{3^n}{3^n+\sqrt{3}}+\frac{\sqrt{3}}{\sqrt{3}+3^n}=1$

$f\left(\frac{1}{100}\right)+f\left(\frac{2}{100}\right).....f\left(\frac{99}{100}\right)$

$\underbrace{f\left(\frac{1}{100}\right)+f\left(\frac{99}{100}\right)}+\underbrace{f\left(\frac{2}{100}\right)+f\left(1-\frac{2}{100}\right)}.....+f\left(\frac{50}{100}\right)$

$1+1+....49\ form+f\left(\frac{1}{2}\right)$

$49+0.5=49.5$

15. Use the properties of log function

Sol: $(\log_e x)^{\log_3 9}=\log_e x-(\log_e x)^2+1$

$\Rightarrow(\log_e x)^2=(\log_e x)-(\log_e x)^2+1$

Let $\log_e x=y$

$2y^2-y-y-1=0\Rightarrow(2y+1)(y-1)=0$

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$y=\frac{-1}{2}$ *and* $y=1$

$\log x=\frac{-1}{2}$ *and* $\log x=1$

But $\log x$ cannot be negative, hence only one solution.

16. $\left|\frac{z-i}{z+2i}\right|=1$

$\Rightarrow |z-i|=|z+2i|$

$\Rightarrow z$ Lies on perpendicular bisector of $(0,1)$ and $(0,-2)$.

$\Rightarrow \text{Im}\, z=-\frac{1}{2}$

Let $z=x-\frac{i}{2}$

$\because |z|=\frac{5}{2}\Rightarrow x^2=6$

$\therefore |z+3i|=\left|x+\frac{5i}{2}\right|=\sqrt{x^2+\frac{25}{4}}$

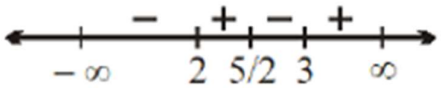
$=\sqrt{6+\frac{25}{4}}=\frac{7}{2}$

17. conceptual

18. Let $f(x)=k(x-2)^2(x-3)^2+3$

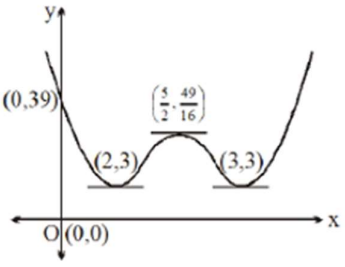
As, $f(1)=4k+3=7$

$\therefore k=1\Rightarrow f(x)=(x-2)^2(x-3)^2+3$



Sign of f ' (x)

Now, $f'(x)=2(x-2)(x-3)(2x-5)$



$\Rightarrow f(x)$ has local maximum at $x=\frac{5}{2}$

$\therefore f_{\max}\left(x=\frac{5}{2}\right)=\frac{49}{16}$

19. $(A->T, B->R, C->S, D->P)$

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$(A)\ a_g = \frac{1}{5} = \frac{a_1 + a_{15}}{2}$

Now sum $= \frac{15}{2}(a_1 + a_{15}) = \frac{15}{2}.\frac{2}{5} = 3$

$(B)\ \frac{1}{y^2 - 4y + 5} < 1 \Rightarrow y^2 - 4y + 5 > 1$

Gives $(y-2)^2 > 0 \Rightarrow y \in R - \{2\}$

$(C)(x+1)^2 = x(2x+2)$

$\Rightarrow x + 1 = 2x \Rightarrow x = 1$

$\therefore$ The sequence is 1,2,4,8

$(D)\ K = \frac{1}{5}\left(\frac{7-2}{2.7} + \frac{12-7}{7.12} + \frac{17-12}{12.17} + ...\right)$

$= \frac{1}{5}\left(\frac{1}{2} - \frac{1}{7} + \frac{1}{7} - \frac{1}{12} + \frac{1}{12} - \frac{1}{17} + ...\right)$

$= \frac{1}{10}$

20. $\omega^{r_1} + \omega^{r_2} + \omega^{r_3} = 0 \left\{1 + \omega + \omega^2 = 0\right\}$

r_1	r_2	r_3
$\downarrow$	$\downarrow$	$\downarrow$
3 *or* 6	1 *or* 4	2 *or* 5

Required prob. $= \frac{2 \times 2 \times 2 \times \angle 3}{6^3} = \frac{2}{9}$

21. All terms are positive

22. $\therefore \sum x_i = 12 \Rightarrow \alpha = 1$

$\alpha = 1$ *and* $\beta = 2$

23. S $D = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & 3 \\ 1 & 2 & a \end{vmatrix} = a$

Unique solution $a \neq 0$

Ordered pair $= 6 \times 7$

24. **$\because (x)$ is continuous at $x = 3$**

$9a + 3b = 20$

$\because fx$ is also differentiable at $x = 3$

$6a + b = 12$

On solving a=3

b = -6

25. $g(x) = \frac{f(x) - f(-x)}{|x| + 2} = \frac{8x^3 - 2}{|x| + 2}$

$\therefore g(x) = g(-x) \quad \therefore g(x)$ is odd.

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PHYSICS

26. $\frac{T_1}{T_2}=\frac{\lambda_2}{\lambda_1}=\frac{350}{510}=0.69$

27. $E=W+ev_o$

$\frac{12400}{3000}evolt=w+ev_1-(1)$

$\frac{12400}{4000}evolt=w+ev_1-(2)$

$equation\,(1)-(2)1.03\,evolt=(v_{o_1}-v_{o_2})e$

$v_{o_1}-v_{o_2}\simeq 1$

28. Conceptual

29. $\mu N\geq mg$

$\mu ma\geq mg$

$a\geq\frac{g}{\mu}$

30. Potential at centre $V_c=K.\frac{3Q}{2R}$

$V_1=K.\frac{Q}{r}$

$K.\frac{3Q}{4R}=\frac{KQ}{r}$

$r=\frac{4R}{3}$

Distance from surface $\frac{4R}{3}-R=\frac{R}{3}$

31. Given, potential energy = Kinetic energy

$\frac{1}{2}m\omega^2x^2=\frac{1}{2}m\omega^2\left(A^2-x^2\right)$

Or $2x^2=A^2;x=\frac{A}{\sqrt{2}}$

32. $\phi=6t^2-5t+1$

$\Rightarrow\xi=-\frac{d\phi}{dt}=-(-12t-5)$

$At\,t=0.25\sec,\xi=2V$

$\Rightarrow I=\frac{\xi}{R}=\frac{2}{10}=0.2A$

33. $W=\frac{\pi r_1r_2}{2}=\frac{\pi\times1\times1}{2}$

$=\pi/2J$

34. $f_w=4\times f_a=4\times12=48cm$

35. conceptual

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36. $\frac{M}{L}=\frac{q}{2m}$

37. conceptual

38. $V_{cm}=\frac{14\times10+0}{14}$

$V_{cm}=m/s$

39. $Y=AB$

40. $\frac{I_{max}}{I_{min}}=\left(\frac{\sqrt{I_1}+\sqrt{I_2}}{\sqrt{I_1}-\sqrt{I_2}}\right)^2=16=4^2$

$\Rightarrow\frac{\sqrt{I_1}+\sqrt{I_2}}{\sqrt{I_1}-\sqrt{I_2}}=4\Rightarrow\sqrt{I_1}+\sqrt{I_2}=4\sqrt{I_1}-4\sqrt{I_2}$

$\Rightarrow5\sqrt{I_2}=3\sqrt{I_1},\frac{I_1}{I_2}=\frac{25}{9}$

41. Conceptual

42. $C'=kC\ and\ so,U'=\frac{1}{2}(kC)v^2=kU.\ also\ q'=C'V$

= kCV = kq and so charge density increases.

43. For an open pipe, the fundamental frequency is $\frac{v}{2l}$

For a closed pipe, the fundamental frequency is $\frac{v}{4l}$

Thus, the frequency of the pipe is twice that of the closed pipe of the same length. Since the are boundary condition (antinode-antinode vs node-antinode) explain this, R correctly explains A

44. Conceptual

45. Conceptual

46. $\frac{B_{center}}{B_{axis}}=\left(1+\frac{x^2}{r^2}\right)^{3/2}\Rightarrow\frac{B_{center}}{54}=\left(1+\left(\frac{4}{3}\right)^2\right)^{3/2}=\frac{125}{27}$

$B_{center}=250\mu T$

47. Diode is reverse bias

48. Bernoulli's equation

49. Newton's law $F=G\frac{m_1m_2}{r^2}$

$G=\frac{Fr^2}{m^2}=\frac{\left[MLT^{-2}\right].\left[L^2\right]}{\left[M^2\right]}=\left[m^{-1}L^3T^{-2}\right]$

$a=-1,b=3,c=-2$

$a+b+c=-1+3-2=0$

50. $\frac{1}{\text{Re}q}=\frac{1}{2R}+\frac{1}{2R}+\frac{1}{R}$

$\text{Re}q=\frac{R}{2}=\frac{10}{2}=5$

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CHEMISTRY

51. NCERT based

52. Silver mirror test is given by aldehyde. Thus formaldehyde and benzaldehyde gives silver mirror test. But formic acid also given positive silver mirror test due to the presence of aldehydic hydrogen group.

53. $K = \frac{x}{t}$(Zero order) $\left(x = [A]_0 - [A]_t\right)$

54. More R_f less its polarity

$$R_f = \frac{Distance\ travelled\ by\ compound\ 'X'}{Distance\ travelled\ by\ \text{solvent}\ 'Y'} = \frac{2}{8} = 0.25 = 25 \times 10^{-2}$$

55. $Hg_2l_2 \rightarrow Hg_2^{+2} + 2l^-$

$i = 1 + 2\alpha \quad [\because \alpha = 1] \qquad i = 1 + 2 = 3$

56. Named reactions

57. Pb

58. NCERT based

59. NCERT based

60.

Element	Atomic mass	Percentage	Moles	Simplest Ratio
C	12	54.2	4.52	2
H	1	9.2	9.2	4
O	16	2.28	2.28	1

Empirical formula $= C_2H_4O$

Molecular formula $= (C_2H_4O)_n$

Empirical formula mass $= 24 + 4 + 16 = 4$

Given molar mass of the compound $135\,g\,mol^{-1}$

$\therefore n = \frac{132}{44} = 3$

$\therefore$ Molecular formula of the compound $= C_6H_{12}O_3$

61. $\overset{+1}{N_2O} < \overset{+2}{NO} < \overset{+3}{N_2O_3} < \overset{+4}{NO_2}$

62. NCERT based

63. Electron affinity order : $Cl > F > O$ Due to strong electron-electron repulsions, the electron affinity of fluorine is less then chlorine.

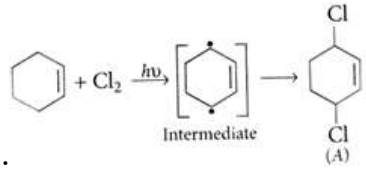
64.

Element	Electronic configuration
^{29}Cu	$[Ar]3d^{10}\ 4s^1$
^{30}Zn	$[Ar]3d^{10}\ 4s^2$
^{48}Cd	$[Kr]4d^{10}\ 5s^2$
^{47}Ag	$[Kr]4d^{10}\ 5s^1$

All elements in option (a) have d^{10} electronic configuration.

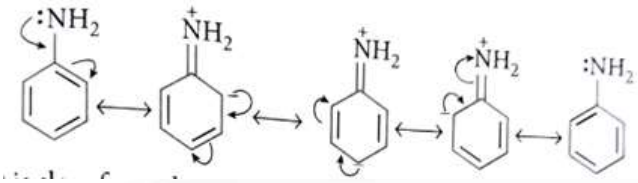
65. $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{+3} + 7H_2O$

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66. 

67. Trigonal Planar

68. Organic Compounds Containing Nitrogen



It is clear from the above resonating structures that the electron density is more in 0 and p positions. Hence, the substitution takes place mainly at these positions. Aniline does not undergo Friedel-Crafts reaction (alkylation and acylation) due to salt formation with aluminium chloride, the Lewise acid, which is used as a catalyst. Due to this nitrogen, of aniline acquire positive charge and hence acts as a strong deactivating group.

69. TACGAAGT

70. K_2HgI_4

71. $N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{3(g)}$

72. **1)** O_2 2) B_2 3) NO

73. Given, $i = 2$

$$MX_3 \rightarrow M^{3+} + 3X^-$$

Before dissociation : 1 0 0

After dissociation : $1-\alpha$ α 3α

$$i = \frac{Total\ no.of\ particles\ after\ dissociation}{Number\ of\ particles\ brfore\ dissociation}$$

$$= \frac{1-\alpha+\alpha+3\alpha}{1} = \frac{1+3\alpha}{14} = 2$$

Percentage dissociation $= 33.33\& \simeq 33\%$

74. 5F

75. 45

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